MIL-I-14925A (PA)
29 March 1974
SUPERSEDING
MIL-I-14925 (MU)
13 December 1968

MILITARY SPECIFICATION

IGNITER ASSEMBLY, M76
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the loading and assembling of one type of igniter designated as Igniter Assembly, M76.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	Powder, Black

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1336

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DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKING DRAWINGS

8799112	Marking Diagram and Sealing for Interplant or CONUS Shipment of Ammunition Fiberboard Boxes
9225157	Case-Launch Motor
9225158	Nozzle-Launch Motor
9225165	Igniter Assembly, M76
9272308	Box, Packing, Ammunition Fiberboard for Igniter Assembly, M76

INSPECTION EQUIPMENT DRAWINGS

9246293	Envelope Drawing Equipment and Instructions for M76 Igniter Assembly Seal Test
9246294	Test Chamber Concept

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Black Powder Moisture Content. The moisture content of the black powder at the time of loading shall not exceed 0.7 percent (%).

3.3 Assembly. The assembly shall comply with the requirements specified on drawing (dwg.) 9225165.

3.4 Igniter Assembly Seals, Airtightness and Strength. The Igniter assembly shall show no evidence of air leakage.

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3.5 Ballistic Functioning. The igniter shall comply with the requirements of Table I.

TABLE I

Ballistic Performance

	-25°F		125°F	
	Min.	Max.	Min.	Max.
a. Fire Pulse to P max., msec	4	26	--	22
b. 0.10 P max. to 0.90 P max., msec	2	13	--	11
c. Maximum Pressure, psig	312	561	288	537

3.6 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. The parts shall be free of cracks, flash, tool marks, blisters, porosity, warpage, dirt and other foreign matter. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.7 First Article Inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all of the inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Submission of Product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. The contractor shall submit a first article sample to a Government approved facility as designated by the Contracting Officer (See 6.4) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 20 igniter bodies (dwg. 9221073) and 30 igniter assemblies (dwg. 9225165) which have been produced by the the contractor or furnished by a supplier, and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production.

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Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials(See 4.1.1c). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawings, specification or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be Performed. Assemblies and components will be subjected by the Government to any or all of the examinations and tests specified in 4.3.2 and 4.3.3 of this specification, and to any or all requirements of the applicable drawings.

4.2.3 Rejection. If any assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection Provisions

4.3.1 Lot Formation. The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn, and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission

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of product as specified in MIL-STD-105. Changes to the process, specification, or drawing, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cover, Igniter (See dwg. 9225162, covering a detail of dwg. 9225165).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:	AQL 1.50 Percent	
201.	Diameter	Gage
202.	Thickness	Gage
203.	Evidence of poor workmanship (See 3.6)	Visual

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4.3.2.2 Body, Igniter, Launch Motor (See dwg. 9221073, covering a detail of dwg. 9225165).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 1.0 Percent	
101.	Contour of nozzle throat mating surface	Gage
102.	Contour of nozzle entrance mating surface	Gage
103.	Diameter of squib cavity	Gage
104.	Wall thickness	Gage
105.	Height of nozzle locking rib	Gage
106.	Land between squib cavities improper	Visual
107.	Surface imperfections	Visual
Minor:	AQL 1.50 Percent	
201.	Diameter of squib wire holes	Gage
202.	Outside diameter, min.	Gage
203.	Evidence of poor workmanship (See 3.6)	Visual

4.3.2.3 Igniter Assembly: M76 (See dwg. 9225165).

Categories	Defects	Method of Inspection
Critical:		
1.	Igniter assembly lead wire shunt improper	Visual
Major:	AQL 0.65 Percent	
101.	Length of insulated wire, min.	Gage
102.	Length of bare wire, min.	Gage
103.	Compliance with note seven	Visual
104.	Bonding of squib to igniter body improper	Visual
105.	Soldering of wire leads improper	Visual
Minor:	AQL 1.50 Percent	
201.	Length of squib tubing	Gage
202.	Type of wire	Visual
203.	Evidence of poor workmanship (See 3.6)	Visual

4.3.2.4 Box, Packing, Ammunition Fiberboard for Igniter Assembly M76 (See dwg. 9272308).

Categories	Defects	Method of Inspection
Critical:	None defined.	

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Major:	AQL 0.65 Percent	
101.	DOT marking missing or incorrect	Visual
102.	Tape missing or installed improperly	Visual
103.	Contents loose (when tilted)	Manual
Minor:	AQL 1.0 Percent	
201.	Marking missing, misleading or incorrect	Visual
202.	Evidence of poor workmanship (See 3.6)	Visual

4.3.3 Testing

4.3.3.1 First Article Sample (See 4.2.1). The assemblies and components submitted as the first article sample shall be tested as specified below after having been examined for requirements as specified in 4.2.2. Rejection and resubmission shall be in accordance with 4.2.3.

4.3.3.1.1 Moisture Content of Black Powder (See 3.2 and 4.3.3.2.1).

4.3.3.1.2 Wall Thickness Variation Examination of Body (See dwg. 9221073 and 4.3.3.2.2) Major Defect. This examination shall be performed on each body using equipment in accordance with 4.3.4.

4.3.3.1.3 Igniter Circuit Resistance (See dwg. 9225165 and 4.3.3.2.3).

4.3.3.1.4 Igniter Assembly Seal (See 3.4 and 4.3.3.2.4).

4.3.3.1.5 Ballistic Functioning (See 3.5). Acceptance for this test shall be in accordance with MIL-STD-414, Section B, Part I for single limits and Section B, Part II for double limits. The sample size and acceptance criteria for each temperature shall be:

Plus (+) 125°F ± Code Letter D, AQL 0.65

Minus (-) 25°F - Code Letter I, AQL 0.65

4.3.3.2 Production Testing

4.3.3.2.1 Moisture Content of Black Powder (See 3.2) Major Defect. A sample shall be selected and tested as specified in MIL-P-223. The test shall be performed at the beginning of each operating shift and each time a new powder storage container is used. Failure to comply with the requirements shall be cause for rejection of the black powder.

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4.3.3.2.2 Wall Thickness Variation Examination of Body (See dwg. 9221073, Major Defect. A sample of 4 bodies from each mold or cavity shall be selected at the end of every 4 hours of production. Failure of any body to comply with the specified requirement shall be cause for rejection of body production represented by the sample. The examination shall be performed using equipment as specified in 4.3.4.

4.3.3.2.3 Igniter Circuit Resistance (See dwg. 9225165) Major Defect. This test shall be performed 100 percent in accordance with 4.4.1, using equipment as specified in 4.3.4. Any assembly which fails to comply with the specified requirement shall be classed defective and shall be removed from the lot.

4.3.3.2.4 Igniter Assembly Seals (See 3.4) Major Defect. This test shall be performed 100 percent in accordance with 4.4.2, using equipment as specified in 4.3.4. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot.

4.3.3.2.5 Ballistic Functioning (See 3.5) Major Defects. At -25°F; a., b., and c. At 125°F, a., b., and c. This test shall be performed following successful completion of igniter assembly seal test in accordance with 4.4.3, using equipment as specified in 4.3.4. Lot acceptance shall be in accordance with the sample size and acceptance criteria of MIL-STD-414, Section B, Part I, for single limits and Section B, Part II for double limits. The sample size and acceptance criteria for each temperature shall be:

Plus (+) 125°F, Code Letter D, AQL 0.65

Minus (-) 25°F, Inspection Level V, AQL 0.65

4.3.4 Inspection Equipment. Inspection equipment required to perform the inspection prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

4.4 Test Methods and Procedures

4.4.1 Igniter Circuit Resistance. This test shall be performed on the igniter assembly without the shorting wire in place. The test

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current shall be limited to 50 milliamperes, maximum. At the completion of the test, the leads shall be shorted.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.4.2 Igniter Assembly Seals. The assembly shall be tested as specified on dwg. 9246293, using equipment as specified on dwg. 9246294.

4.4.3 Ballistic Functioning. The sample igniters shall be installed in nozzles dwg. 9225158 and shall be conditioned at -25°F minus 5°F and +125°F plus 5°F for a period of 4 hours minimum. The nozzle-igniter shall be assembled to a case dwg. 9225157 with a free volume of 23.7 cubic inches. The free volume shall be obtained by installing a simulated propellant in the case with a volume of 20.1 cubic inches. The density of the simulated propellant shall be .0595 pounds per cubic inch. The samples shall be fired within 5 minutes after removal from the conditioning unit. Ballistic results shall be recorded to determine compliance with the requirements. This test shall be performed using a current of 8 ± .5 amperes.

5. PREPARATION FOR DELIVERY

5.1 Packing. Level C. The igniter assembly shall be packed in accordance with dwg. 9272308.

5.2 Marking. The box shall be marked in accordance with dwg. 8799112.

6. NOTES

6.1 Intended Use. The Igniter Assembly, M76 covered by this specification is intended for use in the TOW Launch Motor, M114.

6.2 Ordering Data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

c. Inspection equipment design requirements (See 4.3.4 and 6.3).

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6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3, should he elect to do so.

6.3.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commerical equipment, or with SIE designs if not otherwise covered thereby (See 6.3.1 c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and other specific identifying information from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-R, Dover, New Jersey, 07801, in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of First Article Sample. Instructions as to the submission and location for testing and evaluation of the first article sample shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-I-Q
Dover, New Jersey 07801

6.5 Submission of Test Data. In addition to the normal distribution of records, one copy of all test data shall be forwarded to each of the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-R
and SARPA-AD-E-P-1
Dover, New Jersey 07801

6.6 Definitions

6.6.1 Ballistic Definitions, nomenclature shall be in accordance with MIL-STD-292.

6.6.2 Contractor. Unless otherwise specified, the word "Contractor", used throughout this specification, shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government-Owned, Contractor-Operated (GOCO) and Government-Owned, Government-Operated (GOGO) plants.

Custodian:
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Preparing Activity:
Army-PA

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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